

Finite-volume scalar spectral constructive Gibbs measure – integrated referee package

TECT verification-first repository · claim A4-SCALAR-SPECTRAL-CONSTRUCTIVE-MEASURE

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1. Purpose and scope

This package gives a single-file proof and reproduction map for the finite-volume real-scalar constructive measure in Sector A. It consolidates the previously approved analytic proof and scoped T6 enactment without adding a theorem or enlarging the scope. The torus, regulator, covariance, interaction, topology of convergence, and admissible observables are all fixed below so that the result can be attacked without consulting a sibling note.

The underlying theorem was promoted to T6 after Jusang Lee independently ran the hash-bound verifier on 2026-07-18. Jusang Lee confirmed integrated v2.0 on 2026-07-19, but final adversarial review then found that its shell notation did not isolate the zero mode when $q_0 = 0$. The current v2.1 repairs that endpoint without changing the theorem. It is the exact entry for the eventual reproduction bundle. Jusang Lee confirmed this v2.1 source and PDF on 2026-07-19. That confirmation authorizes bundle-last packaging; publication is claimed only after the resulting bundle passes its integrity and repository release checks.

The result is deliberately scalar. It is not a construction of the full three-complex-component derivative Class-II measure and it proves neither an infinite-volume phase transition nor BCC existence or selection.

1 Definitions and theorem

Let

$$\mathbb{T}_L^3 = [0, L)^3, \quad L = 16, \quad \alpha = \frac{2\pi}{L}, \quad |\mathbb{T}_L^3| = L^3. \quad (2.1)$$

Use the orthonormal real Fourier representation, with the usual conjugacy constraint, and let P_N retain precisely the modes $n \in \mathbb{Z}^3$ satisfying $|n|_\infty \leq N$. For

$$m_{\text{sh}}^2 > 0, \quad Y > 0, \quad q_0 \geq 0, \quad (2.2)$$

define

$$K_n = m_{\text{sh}}^2 + Y(\alpha^2|n|^2 - q_0^2)^2, \quad C_n = K_n^{-1}. \quad (2.3)$$

Let μ_0 be the centred real Gaussian probability measure with covariance $C = K^{-1}$. Its projected law is $\mu_{0,N} = (P_N)_\# \mu_0$.

For $\lambda \in \mathbb{R}$ and $\gamma > 0$, set

$$V(\phi) = \int_{\mathbb{T}_L^3} \left(\frac{\lambda}{4} \phi(x)^4 + \frac{\gamma}{6} \phi(x)^6 \right) dx. \quad (2.4)$$

The finite-dimensional Galerkin Gibbs law is

$$d\nu_N(\phi_N) = Z_N^{-1} e^{-V(\phi_N)} d\mu_{0,N}(\phi_N). \quad (2.5)$$

Theorem (finite-volume scalar spectral constructive measure). Under (2.2) and $\lambda \in \mathbb{R}$, $\gamma > 0$:

1. $C = K^{-1}$ is trace class on real $L^2(\mathbb{T}_L^3)$ and defines a Radon Gaussian probability measure μ_0 ;
2. $V(\phi)$ is finite for μ_0 -almost every ϕ , every Z_N is finite and strictly positive, and the bounds are uniform in N ;
3. with

$$d\nu(\phi) = Z^{-1} e^{-V(\phi)} d\mu_0(\phi), \quad (2.6)$$

the full sequence ν_N converges weakly to ν on real L^2 ;

4. on the common Gaussian space, the lifted normalized densities converge in $L^1(\mu_0)$, equivalently in total variation; and
5. expectations of every finite-degree polynomial in finitely many continuous real linear functionals on L^2 converge.

This is a T6 conditional theorem under the named hypotheses **A1-SHELL-POSITIVITY** and **A2-H2-SEXTIC-COERCIVITY**. The hard analytic dependency is **A1-SCALAR-ANALYTIC-BRANCH**.

2 Trace class and the sharp Gaussian Hilbert threshold

For every integer $m \geq 1$, the number of lattice points on the nonzero max-norm shell $|n|_\infty = m$ is exactly

$$(2m+1)^3 - (2m-1)^3 = 24m^2 + 2. \quad (3.1)$$

Let

$$m_0 = \max \left\{ 1, \left\lceil \sqrt{2}q_0/\alpha \right\rceil \right\}.$$

If $m \geq m_0$ and $|n|_\infty = m$, then $\alpha^2 |n|^2 \geq 2q_0^2$ and

$$K_n \geq \frac{Y}{4} \alpha^4 m^4. \quad (3.2)$$

Therefore, for $M \geq m_0$,

$$\sum_{|n|_\infty \geq M} C_n \leq \frac{4}{Y\alpha^4} \sum_{m=M}^{\infty} (24m^{-2} + 2m^{-4}) < \infty. \quad (3.3)$$

The inner modes are finite in number and $K_n \geq m_{\text{sh}}^2$, so $\text{Tr } C < \infty$. Hence μ_0 is a Radon Gaussian measure on L^2 .

With the H^s weight, the shell contribution is asymptotic to m^{2s-2} . Thus the covariance is trace class on H^s for every $s < 1/2$. Conversely, the upper estimate $K_n \leq c(1 + |n|^4)$ at large $|n|$ gives the harmonic lower behaviour at $s = 1/2$, so the threshold is sharp. This regularity fact is used as a warning, not as a Sobolev route to the local sextic term.

3 The local interaction: direct Gaussian L^6 control

Let $X_N = (I - P_N)\phi$. Translation invariance gives the pointwise variance

$$t_N = |\mathbb{T}_L^3|^{-1} \sum_{|n|_\infty > N} C_n \longrightarrow 0. \quad (4.1)$$

For a centred real Gaussian random variable X , $\mathbb{E}X^6 = 15(\mathbb{E}X^2)^3$. Tonelli's theorem therefore yields

$$\mathbb{E} \int_{\mathbb{T}_L^3} |X_N(x)|^6 dx = 15 |\mathbb{T}_L^3| t_N^3 \longrightarrow 0. \quad (4.2)$$

Consequently $P_N \phi \rightarrow \phi$ in $L^6(\Omega \times \mathbb{T}_L^3)$, $\phi \in L^6$ almost surely, and the quartic and sextic integrals in (2.4) are finite. This is the load-bearing construction of V ; the false embedding $H^{1/2-} \hookrightarrow L^6$ in three dimensions is never used.

The polynomial difference identity and Holder's inequality on the finite measure product space imply

$$\mathbb{E}|V(P_N \phi) - V(\phi)| \longrightarrow 0. \quad (4.3)$$

For the sextic part one uses

$$|a^6 - b^6| \leq 6(|a|^5 + |b|^5)|a - b| \quad (4.4)$$

with the L^6 convergence and uniform Gaussian L^6 norms; the quartic term follows identically with lower powers.

4 Stability and uniform partition bounds

Put $a = \max(-\lambda, 0)$ and $x = \phi^2$. Direct minimization on $x \geq 0$ gives

$$\frac{\lambda}{4}\phi^4 + \frac{\gamma}{6}\phi^6 \geq -\frac{a^3}{12\gamma^2} =: -c_*. \quad (5.1)$$

Therefore

$$0 < e^{-V(P_N \phi)} \leq e^{c_* |\mathbb{T}_L^3|}, \quad Z_N \leq e^{c_* |\mathbb{T}_L^3|}. \quad (5.2)$$

Let $\sigma_*^2 = |\mathbb{T}_L^3|^{-1} \text{Tr } C$ and $\lambda_+ = \max(\lambda, 0)$. The real Gaussian moments $\mathbb{E}X^4 = 3\sigma^4$ and $\mathbb{E}X^6 = 15\sigma^6$ give

$$\mathbb{E}V(P_N \phi) \leq |\mathbb{T}_L^3| \left(\frac{3\lambda_+}{4} \sigma_*^4 + \frac{5\gamma}{2} \sigma_*^6 \right) =: B_* < \infty. \quad (5.3)$$

Since $x \mapsto e^{-x}$ is convex, Jensen's inequality has the direction

$$Z_N = \mathbb{E}e^{-V(P_N \phi)} \geq e^{-\mathbb{E}V(P_N \phi)} \geq e^{-B_*} > 0. \quad (5.4)$$

Equations (5.2)–(5.4) enclose every Galerkin partition function uniformly.

5 Full-sequence cutoff removal

Write $w_N = e^{-V(P_N \phi)}$ and $w = e^{-V(\phi)}$. Equation (4.3) implies $V(P_N \phi) \rightarrow V(\phi)$ in probability. The deterministic ceiling in (5.2) makes the weights uniformly bounded, and bounded convergence in probability gives

$$w_N \longrightarrow w \quad \text{in } L^1(\mu_0), \quad Z_N = \mathbb{E}w_N \longrightarrow Z = \mathbb{E}w > 0. \quad (6.1)$$

Define the lifted law

$$d\tilde{\nu}_N = Z_N^{-1} w_N d\mu_0. \quad (6.2)$$

Then

$$\left\| \frac{w_N}{Z_N} - \frac{w}{Z} \right\|_{L^1(\mu_0)} \longrightarrow 0. \quad (6.3)$$

This is total-variation convergence only on the common Gaussian space.

The actual Galerkin law satisfies $\nu_N = (P_N)_\# \tilde{\nu}_N$. For bounded continuous F on L^2 , $P_N \phi \rightarrow \phi$ almost surely and (5.2)–(5.4) uniformly bound the normalized densities. Combining this fact with (6.3) gives

$$\int F(\phi_N) d\nu_N = \int F(P_N \phi) d\tilde{\nu}_N \longrightarrow \int F(\phi) d\nu. \quad (6.4)$$

Thus the full sequence converges weakly; no unnamed subsequence remains. The embedded finite-dimensional laws have different supports and are not claimed to converge in total variation.

6 Declared correlations

Let $\ell_1, \dots, \ell_r$ be continuous real linear functionals on L^2 and Q a finite-degree polynomial. The random variables $Q(\ell_1(P_N \phi), \dots, \ell_r(P_N \phi))$ converge in every finite Gaussian moment to the corresponding unprojected polynomial. The uniform density ceiling following from (5.2)–(5.4) transfers uniform integrability to the lifted Gibbs laws. Hence

$$\int Q(\ell_1(\phi_N), \dots, \ell_r(\phi_N)) d\nu_N \longrightarrow \int Q(\ell_1(\phi), \dots, \ell_r(\phi)) d\nu. \quad (7.1)$$

Unsmearing coincident-point products and composite-operator renormalization are not included.

7 Reproduction and independent execution

The one-command verifier is

```
python codes/foundations/a4_scalar_constructive_measure_verify.py
```

and must print

```
PASS: primary (18/18)
PASS: independent (15/15)
ASSERTS: 33/33
A4-SCALAR-CONSTRUCTIVE-INTEGRATED-PASS
```

The primary audit derives the trace, stability, partition, and Gaussian-tail bounds. The second route is non-importing: it reconstructs the shell count, Fourier traces, Decimal potential minimum, Gaussian moments, and partition bounds with scalar loops. The integrated verifier launches both routes in a fresh temporary directory and checks the three source hashes pinned in `constructive_measure_manifest.json`.

Jusang Lee independently executed verifier v1.0.0 from clean commit

```
7eee2fe84887cc4ccdd311c75095ec84bc9d0d45
```

on 2026-07-18. The immutable operator artifact is

```
claims/A4-SCALAR-SPECTRAL-CONSTRUCTIVE-MEASURE/runs/
2026-07-18-jusang-independent/result.json
```

with SHA-256

```
fb2d81ba75d95972e55c675146fbf84cb8209847ea031abdaa0a5f44b111935c
```

and 31/31 assertions, zero failures, zero bad source rows, and zero bad audit rows. Verifier v1.1.0 changed no mathematics or audit calculation; it removed only the stale pre-promotion boundary sentence. Final review of v2.0 then identified the $q_0 = 0$ shell-index endpoint described above. Primary and independent audit v1.1.0 add separate boundary assertions, and verifier v1.2.0 reruns the full package. Its v2.1 referee preflight passes 33/33, at

claims/A4-SCALAR-SPECTRAL-CONSTRUCTIVE-MEASURE/runs/
2026-07-19-referee-preflight-v2.1/result.json

with SHA-256

85da0df0d2b96dbfc98f2ea8a0787bf1bd711228505c671c06cb9d3e036836d8.

The deterministic internal and operator JSONs may have identical content hashes; execution independence is instead evidenced by the clean source commit, separate operator-created path and filesystem event, and fresh subprocess return codes. Algorithmic diversity comes from the non-importing second route.

8 Quantitative sanity record

At cutoff 8 the scalar-loop and vectorized covariance traces agree to at most 1.32×10^{-16} relative error. The production-local shell mass is reconstructed from its stored inputs as

$$r - \frac{Z^2}{4Y} = 0.260000000009475, \quad (9.1)$$

consistent with the stored value 0.26 at the declared source precision. The conservative covariance-trace bounds are 3029.399163006103 for the perturbative scalar anchor and 596.857766699839 for the production-local anchor. The pinned-coefficient pointwise stability constant is 0.0025246087994716246 per unit volume. These numbers are executable sanity checks, not physical predictions or practical cutoff error estimates.

9 Devil’s-advocate review

1. **Trace class alone defines the local sextic. UPHOLD as false.** The separate Gaussian sixth-moment estimate (4.2) is load-bearing.
2. **The proof silently uses $H^{1/2-} \hookrightarrow L^6$. UPHOLD as false.** That embedding is unavailable in three dimensions; the proof uses direct $L^6(\Omega \times \mathbb{T}^3)$ convergence.
3. **A negative quartic makes the Gibbs weight uncontrollable. DISMISSED under the hypotheses.** The positive sextic gives the exact global lower bound (5.1).
4. **Jensen’s inequality has the wrong direction. DISMISSED.** Convexity of e^{-x} gives the lower bound (5.4).
5. **Tightness identifies only a subsequence. UPHOLD as insufficient by itself.** The common-space L^1 density limit identifies the full sequence before weak convergence is concluded.
6. **The finite-dimensional laws converge in total variation after zero extension. UPHOLD as false.** Total variation is asserted only for the lifted laws; the projected laws converge weakly.
7. **Identical deterministic JSON hashes prove a copied operator artifact. DISMISSED.** The execution provenance and subprocess returns are distinct; a non-importing second algorithm supplies diversity.
8. **The theorem constructs derivative Class-II currents. UPHOLD as false.** Typical Gaussian regularity does not automatically define those terms; they require a separate construction or renormalization.

9. **Finite volume proves a phase transition or BCC selection. UPHELD as false.** Neither a thermodynamic limit nor a selection argument occurs here.
10. **The large explicit constants are usable numerical error bars. UPHELD as false.** They certify integrability and reproduction only.
11. **The shell-count proof is undefined at $q_0 = 0$. DISMISSED after v2.1 repair.** The zero Fourier mode is part of the finite inner set, while the tail begins at $m_0 = \max\{1, \lceil \sqrt{2q_0/\alpha} \rceil\}$; two executable routes test the $q_0 = 0, m_0 = 1$ endpoint independently.

10 Falsifier and honesty boundary

The theorem fails if the covariance trace diverges in the declared positive q^4 scope; the Gaussian L^6 tail does not vanish; the exact stability or uniform partition bounds fail; the common-space densities do not converge in L^1 ; two cutoff subsequences have different weak limits; a declared smeared cylinder polynomial fails to converge; a source/evidence hash drifts; or one of the two executable routes fails.

The theorem excludes the full three-component derivative Class-II Gibbs measure, $\eta_{\text{shell}} \neq 0$, zero or negative shell mass, unsmeared composite operators, infinite volume, phase transition, finite-difference Route B and regulator independence, BCC existence or selection, T7, and any claim that the scalar branch shares the full-production parameter anchor.

Result footer

Result ID:	A4-SCALAR-SPECTRAL-CONSTRUCTIVE-MEASURE
Precise statement:	Full sharp spectral Galerkin Gibbs sequence converges weakly on real L2; lifted densities converge in L1/TV; finite-degree smeared cylinder polynomial correlations converge.
Scope:	T6 CONDITIONAL-THEOREM@FINITE-VOLUME-REAL-SCALAR-SPECTRAL; L=16, m_sh^2>0, Y>0, q0>=0, lambda real, gamma>0, beta=1.
Dependencies:	A1-SCALAR-ANALYTIC-BRANCH; named hypotheses A1-SHELL-POSITIVITY and A2-H2-SEXTIC-COERCIVITY.
Evidence grade:	ANALYTIC + EXACT + EXECUTED + CONDITIONAL.
Reproduction command:	python codes/foundations/a4_scalar_constructive_measure_verify.py
Expected output:	primary 18/18; independent 15/15; total 33/33; A4-SCALAR-CONSTRUCTIVE-INTEGRATED-PASS.
Operator evidence:	2026-07-18-jusang-independent/result.json; SHA-256 fb2d81ba75d95972e55c675146fbf84c...
Publication preflight:	2026-07-19-referee-preflight-v2.1/result.json; SHA-256 85da0df0d2b96dbfc98f2ea8a0787bf1...
Falsification gate:	Failed proof step, route, source/evidence hash, or execution provenance in the declared scope.
Tier:	T6 CONDITIONAL-THEOREM, operator-approved 2026-07-18.
Package status:	exact v2.1 entry document operator-confirmed by Jusang Lee on 2026-07-19 after the q0=0 repair; PUBLISHED bundle-last packaging authorized.
No-overclaim statement:	Not derivative Class-II, unsmeared composites, infinite volume, phase transition, Route B, BCC, T7, or parameter identity with the full branch.
Next required action:	Build, integrity-check, and register the T6 PUBLISHED bundle before the A5 capstone bundle.